

B105 Applied Statistical Modelling

Statistical Analysis of Customer Revenue: Germany and the United Kingdom

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Dataset: UCI Online Retail

Dataset URL: <https://archive.ics.uci.edu/dataset/352/online+retail>

GitHub Repository <https://github.com/Sunveerkaur22/B105-applied-statistical-modelling>

1. Introduction

The data analysed in this project is transaction-level sales data from the UCI Online Retail dataset. The dataset comprises transaction-level information from a UK-based online retailer including invoice number, product code, product name, quantity sold, invoice date, unit price, customer ID and country.

The focus of the business in this project is customer revenue from Germany and the United Kingdom. Knowing if there is a difference in revenue generated in these countries would give useful insight into how well the customers perform internationally and where more research may be needed.

In order to conduct the statistical test, data at the transaction level was gathered; however, for the actual statistical comparison customer-level data will be used. This way we avoid counting as two independent observations transactions belonging to the same customer. Revenue was calculated by multiplying the quantity of products purchased by the unit price, and then log transformation was used since revenue values were skewed to the right in the original dataset.

2. Business Question and Hypotheses

Research **Question:** Is customer-level revenue significantly different between customers in Germany and the United Kingdom?

H0: Mean log customer revenue is equal in Germany and the United Kingdom

H1: Mean log customer revenue is different in Germany and the United Kingdom. Alpha = 0.05 (5%).

3. Data Preparation and Cleaning

The original UCI Online Retail dataset included 541,909 transactions. Data was loaded in R using the readxl package. Original data is stored in retail, while new cleaned dataset is called sales.

C-invoice numbers were regarded as a cancellation transaction and therefore ignored. A total of 9,288 C-invoices were identified. A-invoice numbers represented accounting/bad debt adjustments and were therefore ignored.

Records with non-positive quantities were ignored since the analysis was based on positive sales revenue. Non-positive unit price records were also ignored in the calculation of positive sales revenue. Revenue was determined by multiplication of Quantity by Unit Price. The total number of records in the cleaned sales data was 530,103.

```

71 library(dplyr)
72 # Create cleaned sales dataset
73
74 sales <- retail %>%
75   filter(
76     !grepl("^C", InvoiceNo),
77     !grepl("^A", InvoiceNo),
78     Quantity > 0,
79     UnitPrice > 0
80   )
81 # Check cleaned dataset

```

In the main analysis, only Germany and the UK countries were selected. Transactions that did not have CustomerID were ignored in the customer-level aggregation due to the impossibility of allocation of revenue.

No random sample was drawn. All eligible customers in the selected countries were retained. The final customer-level dataset contained 4,014 customers: 94 German customers and 3,920 UK customers.

4. Exploratory Data Analysis

Customer-level total revenue was calculated by summing transaction revenue for each customer. The descriptive statistics are shown below

Descriptive statistics Germany: 94 customers, mean revenue £2,435, median £1,066, SD £3,268, mean log revenue 7.11. United Kingdom: 3,920 customers, mean revenue £1,864, median £652, SD £7,483, mean log revenue 6.54.

A tibble: 2 × 6

	Country	Customers	Mean_Revenue	Median_Revenue	SD_Revenue	Mean_Log_Revenue
1	Germany	94	2435.	1066.	3268.	7.11
2	United Kingdom	3920	1864.	652.	7483.	6.54

The difference between the means and medians indicates substantial skew and the influence of high-revenue customers. The standard deviations also show considerable variation, particularly in the UK group.

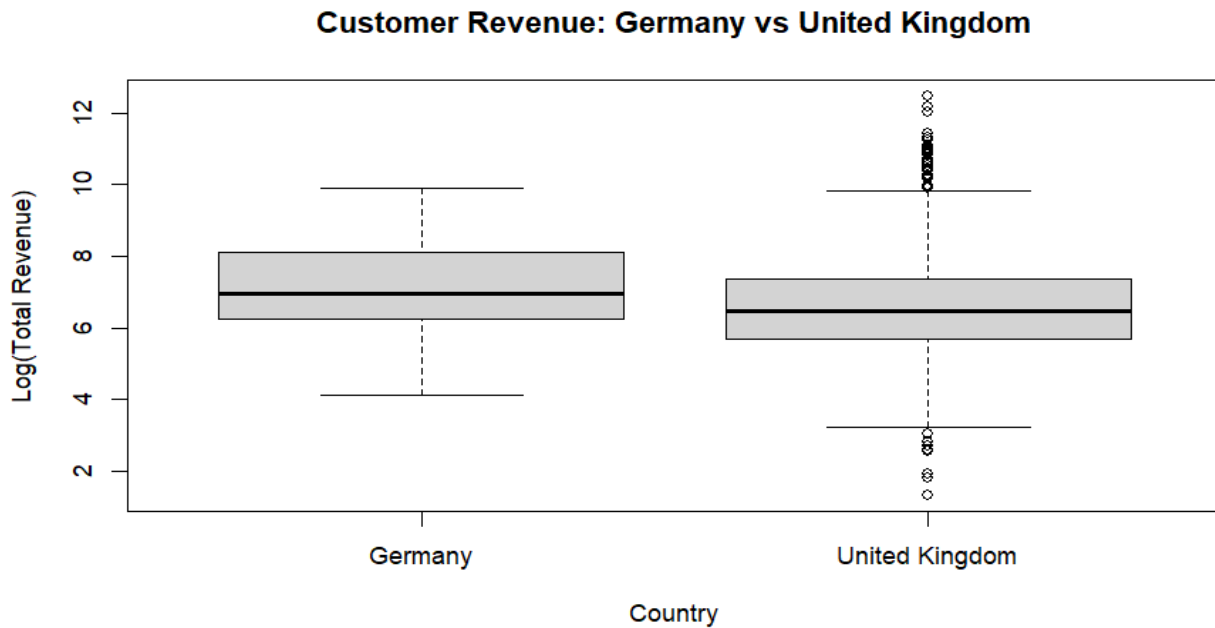


Figure 1. Boxplot of log-transformed customer revenue for Germany and the United Kingdom.

4.1 Log Transformation and Normality

A natural logarithmic transformation was applied using $\text{LogRevenue} = \log(\text{TotalRevenue})$. The transformation reduced the effect of extreme revenue values and produced a distribution more suitable for comparison.

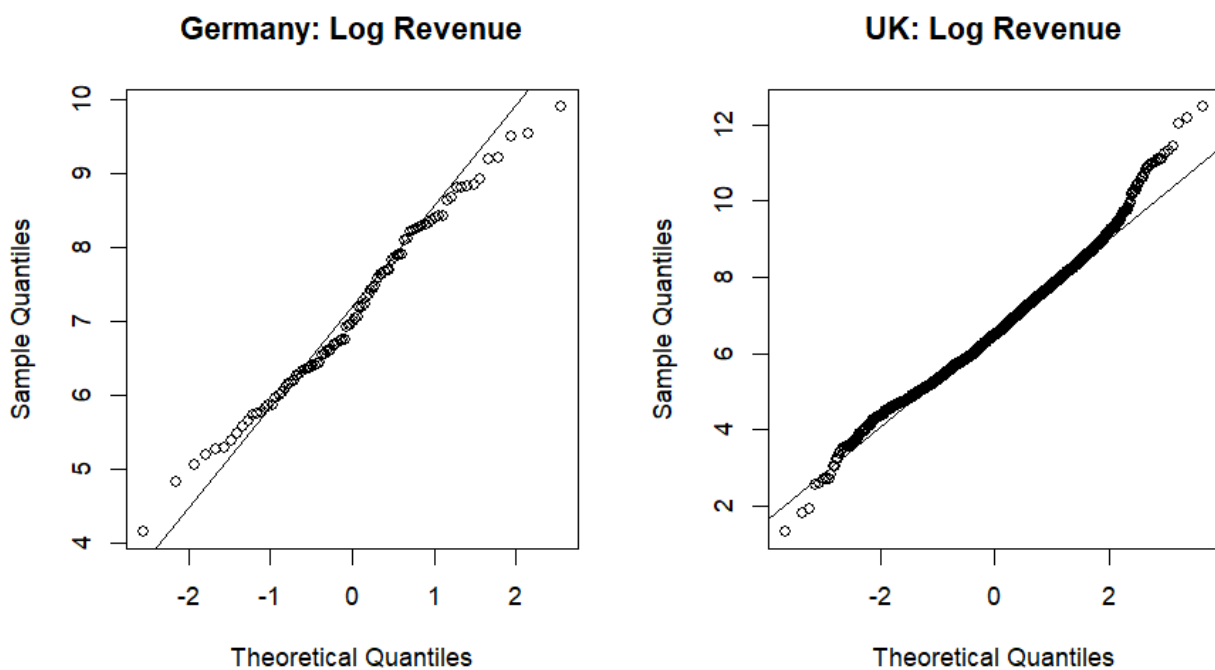


Figure 2. Q-Q plot of log-transformed customer revenue.

For Germany, the Shapiro-Wilk test gave $W = 0.9872$ and $p = 0.4968$. For the United Kingdom, $W = 0.99033$ and p approximately 1.04×10^{-15} .

The German statistic does not indicate that there is evidence of a departure from normality at the 5% significance level. The UK statistic indicates that there is statistical evidence of departure from normality at the 5% level. Normality can therefore not be said to be completely satisfied; there is some improvement in distribution but not completely normality.

5. Inferential Statistical Analysis

A Welch two-sample t-test was performed to check whether mean log customer revenue differs between Germany and the United Kingdom. It is important to use Welch's test since, according to descriptive statistics, the samples are characterized by different variabilities. Additionally, customers should be used as the observational units rather than transactions.

5.1 R Test Code.

The code was:

```
> # Welch two-sample t-test
>
> test_result <- t.test(
+   LogRevenue ~ Country,
+   data = customer_revenue,
+   var.equal = FALSE
+ )
>
> test_result

Welch Two Sample t-test

data:  LogRevenue by Country
t = 4.4666, df = 97.833, p-value = 2.134e-05
alternative hypothesis: true difference in means between group Germany and group United Kingdom is not equal to 0
95 percent confidence interval:
 0.3125520 0.8123384
sample estimates:
mean in group Germany mean in group United Kingdom
       7.105649              6.543204
```

5.2 Test Results

The results were:

Germany mean log revenue = 7.105649

UK mean log revenue = 6.543204

Difference = 0.562445 $t = 4.4666$

Degrees of freedom = 97.833 p -

value = 2.134×10^{-5}

95% CI for the log difference = 0.312552 to 0.812338

Since the p -value is less than 0.05, the finding is statistically significant. The estimated difference in mean log customer revenue is approximately equal to 0.562. After exponentiation, the ratio of means equals approximately 1.755. The corresponding 95% confidence interval was approximately

1.367 to 2.253.

Hence, in terms of this sample, the geometric mean customer revenue in Germany is estimated to be approximately 1.75 times larger than the same mean for the United Kingdom. It is an association between country and customer revenue, but not a causal relationship.

6. Business Interpretation

In sum, the statistical analysis detects a statistically detectable difference in customer-level revenue between Germany and the United Kingdom. Descriptive statistics indicate a difference in the mean and median customer revenue between the two countries. Additionally, inferential statistics performed on log-transformed revenue detects a difference as well.

The ratio of geometric mean revenues of approximately 1.75 is the interpretable parameter, and the confidence interval of approximately 1.37 to 2.25 shows uncertainty.

From a business perspective, the retailer may analyze why there are differences in customer revenue in each market. Factors that may include purchase frequency, products mix, average basket size, customers and marketing activities, but this analysis doesn't show the reason or causal connection.

7. Limitations

The dataset consists of historical transactions from a certain time period and may not represent the current behavior of customers. Most of the transactions do not have CustomerID and thus cannot be linked to any individual customer.

Non-positive quantities and unit prices were excluded from the positive-sales revenue analysis. Some zero-price records could potentially represent legitimate promotional or free items.

Customer revenue is very variable and is still not normally distributed even after transformation, especially in the case of UK group. Also, the number of customers is different for each group, since there are 94 German customers while there are 3,920 in the UK group.

Finally, country is considered as the main grouping variable, so the analysis cannot establish causal explanations for the observed difference.

8. Future Work

Future work might involve looking at purchase frequency, average order value and product categories to see whether those will explain the country difference. A more general research including other countries would identify whether such a pattern happens just for Germany and the UK or not. A more complex statistical model would allow including several explanatory variables at once instead of considering the country only.

9. Conclusion

This project looked into whether there was any difference in customer revenue between Germany and the United Kingdom using UCI Online Retail data. After cleaning and aggregation to the customer level, the sample included 94 German customers and 3,920 UK customers. There was a strong skewness in revenue, and a log-transform was used.

According to Welch two-sample t-test there is a statistically significant difference between mean log customer revenue, $t = 4.4666$, $p = 2.134 \times 10^{-5}$. The difference is estimated as 0.562, which means that the estimated geometric mean ratio of revenues is about 1.75, with a 95% confidence interval of 1.37-2.25.

The results suggest that there is a difference in customer revenue between the two countries in the dataset. Nevertheless, it is necessary to consider distribution characteristics, the lack of customer IDs, differences in group size, and the fact that the analysis was not causal.

10. References

UCI Machine Learning Repository (n.d.) *Online Retail*. Available at:
<https://archive.ics.uci.edu/dataset/352/online+retail> (Accessed: [15-09-2026])..

11. GitHub Repository

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